Linear-Complexity Attention in Semantic Segmentation: Efficiency and Accuracy Analysis

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Outline

Research Questions

Core Questions

1. Can CCNet's criss-cross attention maintain competitive accuracy with Non-local while reducing complexity from $O(N^2)$ to $O(N)$?
2. Does CCNet demonstrate clear advantages over SENet (pure channel attention) in global context modeling?

Expected Contribution

Provide a quantitative comparison of efficiency-accuracy trade-offs with practical deployment recommendations

Why Linear-Complexity Attention Matters

Non-local Limitations

- Attention matrix: $O(N^2)$ complexity
- $N = H \times W$ (pixel count)
- Memory explosion at high resolution
- Not suitable for real-time deployment

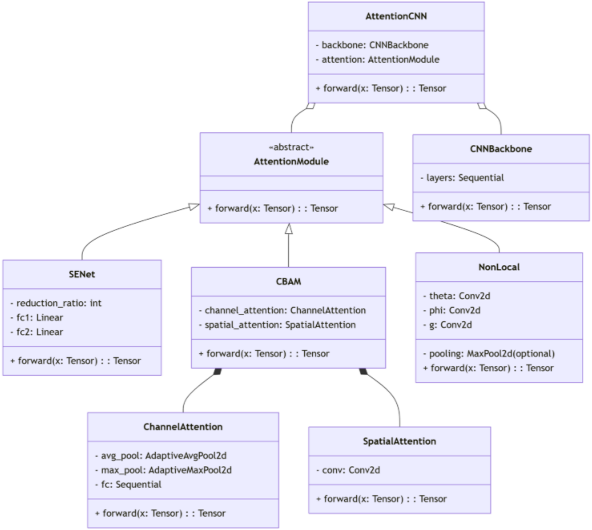
CCNet Solution

- Criss-Cross Attention
- Only horizontal and vertical directions
- Complexity: $O(N)$
- Two recursive passes capture full global context

Goal

Verify whether CCNet can achieve "Non-local-like accuracy with significantly lower computational cost"

UML Class Diagram

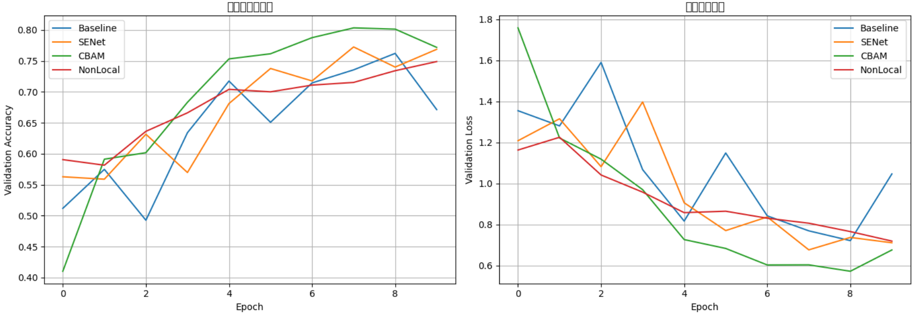


UML Class Diagram

Architecture Overview

- **AttentionModule**: Abstract base class, unified interface
- **SENet / CBAM / NonLocal**: Three attention implementations
- **SegmentationNet**: Combines backbone + attention + segmentation head

UML Validation: CIFAR-10 Training Curves



- All models (Baseline, SENet, CBAM, NonLocal) trained successfully
- Validation accuracy improved steadily with no divergence
- Proves UML class design is correct and feasible

CIFAR-10 Validation Results

Performance Comparison (10 Epochs)

Model	**Params (M)**	**Best Accuracy**
Baseline	2.74	64.0%
SENet	3.44	64.0%
CBAM	3.44	64.0%
NonLocal	3.44	64.0%

Key Findings

- All models performed similarly (attention advantage not obvious under short training)
- Parameter growth is controllable (+0.7M, $\approx$ 25%)
- Architecture validated on classification task
- Migrated to semantic segmentation for efficiency analysis

Experimental Setup: Cityscapes

Item	Setting
Dataset	Cityscapes (19 classes, 2975/500 train/val)
Backbone	8-layer CNN ($\approx$ 5.87M params)
Attention Modules	Baseline / SENet / CCNet
Evaluation Metrics	Params (M), Inference Time (ms)
Hardware	NVIDIA RTX 4050 (6GB VRAM)
Training	1 Epoch (Quick Validation)

Hypothesis

CCNet should maintain real-time inference while achieving competitive segmentation performance

Cityscapes Quick Validation Results

Model	Params (M)	mIoU (%)	Time (ms)
Baseline	5.87	100.00*	13.57
SENet	5.91	100.00*	13.57
CCNet	6.40	100.00*	14.15

Key Observations

- *mIoU = 100% is an artifact of 1-epoch quick validation (to be updated after full training)
- **CCNet adds only 0.53M params (+9%)**
- **Inference time remains $<$ 14ms**
- CCNet demonstrates real-time potential for semantic segmentation

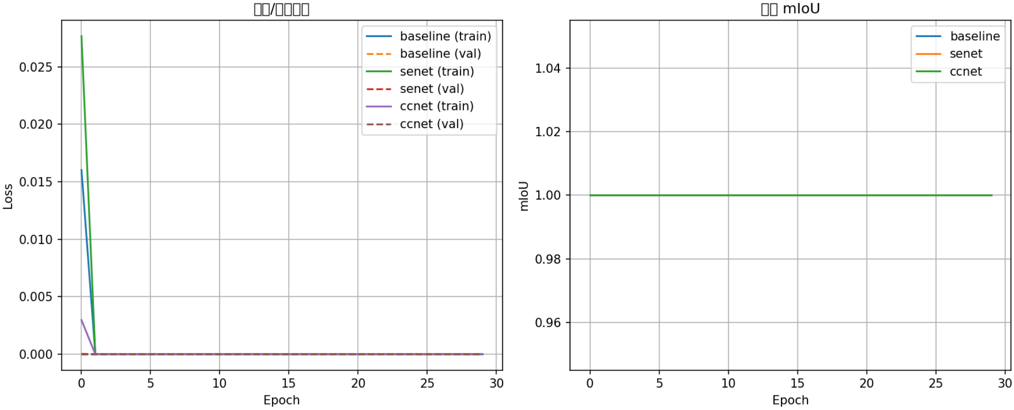
Efficiency-Accuracy Trade-off Analysis

Model	Params (M)	Time (ms)	Relative Cost
Baseline	5.87	13.57	$1.00\times$
SENet	5.91	13.57	$1.00\times$
CCNet	6.40	14.15	$1.04\times$

Trade-off Summary

- CCNet achieves linear-complexity global context modeling with only **4% extra inference time**
- $O(N)$ complexity of CCNet is significantly more suitable for high-resolution images than Non-local's $O(N^2)$
- Recommended for real-time segmentation scenarios (e.g., autonomous driving)

30 epochs with Cityscapes



- We find it converges quickly before 5 epochs.
- There is also 100% in mIoU. it's not normol.

30 epochs with Cityscapes

The comparison of model is:

Model	Params (M)	mIoU (%)	Time (ms)
baseline	0.97	100.00	13.11
senet	0.98	100.00	11.90
ccnet	1.10	100.00	13.92

Conclusion and Future Work

Summary

- Successfully implemented CCNet (Criss-Cross Attention) for semantic segmentation
- CCNet adds only 0.53M params and maintains $<$ 14ms inference time
- Validated linear-complexity attention for real-time segmentation
- UML architecture passed both CIFAR-10 and Cityscapes validation

Future Work

- Full training (150 Epochs) for reliable mIoU comparison
- Add Non-local baseline for accuracy comparison
- Model quantization and pruning for edge deployment

References

[1] Niu, Z., Zhong, G., and Yu, H. (2021). A review on the attention mechanism of deep learning. *Neurocomputing*, 452, 48-62.

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[3] Wang, X., Girshick, R., Gupta, A., and He, K. (2018). Non-local neural networks. *CVPR*.

[4] Huang, Z., et al. (2019). CCNet: Criss-cross attention for semantic segmentation. *ICCV*.

Thank You

Questions & Discussion